

Qian Gao

Date of Birth: 09-05-1997

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Present address: School of Electronic Engineering and Computer Science, Queen Mary University of London, London E1 4NS, UK



Scholastics

- 2022.09-2026.09 Queen Mary University of London Doctor
Supervisor: Prof. Yuanwei.Liu, Fellow, IEEE
Major: Electronic Engineering
Research Interest: WiFi human sensing, integrated sensing and communication, machine learning.
- 2019.09-2022.06 Beijing University of Posts and Telecommunications (211) Master
Major: Information and Communication Engineering (A+)
Research Project: The algorithm improvement of Reinforcement Learning mainly in value-based areas and its validation based on Atari games.
GPA:86.57/100
- 2015.09-2019.07 Harbin Engineering University (211) Bachelor
Major: Electronic Information Engineering
Research Project: Calibrate the acoustic center of the sonar array by using the Gauss-Newton iterative method.
GPA:85.36/100

Publish (<https://scholar.google.com/citations?user=PXR6luYAAAAJ&hl=zh-CN>)

Q.Gao, R. Zhong, H. Shin and Y. Liu, "LLM-enabled Antenna Partitioning and Beamforming Optimization for Segmented Pinching Antenna assisted ISAC Systems," in IEEE Transactions on Wireless Communication, submitted, 2026.

Q.Gao, R. Zhong and Y. Liu, "Graph-Enhanced LLM for SWAN-ISAC," in GLOBECOM 2026 - 2026 IEEE Global Communications Conference, submitted, 2026.

Q.Gao, R. Zhong, H. Shin and Y. Liu, "Integrated Sensing and Communication for Segmented Waveguide-Enabled Pinching Antenna Systems," in IEEE Transactions on Vehicular Technology, under revision, 2026.

Q.Gao, R. Zhong, Y. Liu, H. Shin and Y. Liu, "RL based Beamforming Optimization for 3D Pinching Antenna assisted ISAC Systems," in IEEE Transactions on Vehicular Technology, accepted, 2026.

Q. Gao, R. Zhong and Y. Liu, "Deep Learning Based Three-Stage Solution for ISAC Beamforming Optimization," GLOBECOM 2025 - 2025 IEEE Global Communications Conference, accepted, 2025.

Q. Gao, R. Zhong, Y Zou, H. Shin and Y. Liu, "Deep Learning based Beamforming Optimization for ISAC Systems: A Low-complexity and Transferable Framework," in IEEE Transactions on Wireless Communication, pp. 8754 - 8768.

Q. Gao, R. Zhong, X. Mu, M. Peng, H. Shin and Y. Liu, "Distributed Antenna Selection and Beamforming for ELAA based Downlink ISAC Systems," in IEEE Transactions on Vehicular Technology, pp. 17865 - 17878.

Q. Gao, R. Zhong and Y. Liu, "Trajectory and Beamforming Optimization in UAV-enabled ISAC System," GLOBECOM 2024 - 2024 IEEE Global Communications Conference, Cape Town, South Africa, 2024, pp. 1527-1532.

Q. Gao, R. Zhong and Y. Liu, "Antenna Selection and Beamforming in ELAA-based ISAC System: A Distributed Approach," GLOBECOM 2024 - 2024 IEEE Global Communications Conference, Cape Town, South Africa, 2024, pp. 1979-1984.

Q. Gao, R. Zhong, H. Shin and Y. Liu, "MARL-Based UAV Trajectory and Beamforming Optimization for ISAC System," in IEEE Internet of Things Journal, vol. 11, no. 24, pp. 40492-40505, 15 Dec.15, 2024.

Q. Gao, Y. Hao and Y. Liu, "AutoSen: Improving Automatic WiFi Human Sensing through Cross-Modal Autoencoder," ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Korea, Republic of, 2024, pp. 8235-8239.

Q. Gao, Y. Zhang and Y. Liu, "Fuzzy Noisy Network for Stable Exploration," 2021 IEEE 21st International Conference on Communication Technology (ICCT), Tianjin, China, 2021, pp. 792-796

Honor

- 2024 QMUL Postgraduate Research Fund
- 2022 QMUL-CSC Joint Scholarship
- 2022 Outstanding graduate of BUPT
- 2021 National Scholarship for Graduate Students
- Excellent Oral Presentation Award in ICCT Conference, Tianjin

- 2020 iTeach competition, National First Price
- The 17th China Graduate Mathematical Modeling, National Third Price